

PERSONAL PROFILE



Name Lai Nguyen Huy
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EDUCATIONAL BACKGROUND

1. Doctor of Eng. Environmental Technology and Management, Asian Institute of Technology, Pathumthani, Thailand; Jan 2018 – Dec 2021.
2. Master. Eng. Environmental Technology and Management, Asian Institute of Technology, Pathumthani, Thailand; Aug 2013 – May 2015. Outstanding prize (3.83/4.00)
3. Bachelor. Eng. International University – Vietnam National University, Ho Chi Minh City, Vietnam; Aug 2008 – Nov 2012. Second class honor (79.3/100)

LANGUAGES

Vietnamese (mother tongue); English (good)

EMPLOYMENT DETAILS

Date	Name of Organization
Jan 2022 to Present	Research Specialist at Asian Institute of Technology, Thailand
Aug 2021 to Dec 2021	Senior Researcher at Clean Air Asia, Philippines (Headquarter)
Jan 2018 to Aug 2021	Assistant for projects and student mentoring.
Aug 2016 until Oct, 2017	Research Associate at the Hong Kong Polytechnic University, Hong Kong.
Jun 2015 until Jun, 2016	Research Associate at EEM/SERD/AIT, Thailand
Jun-Jul, 2014	Internship at K-water Cooperation (Daejeon, Korea)
Dec 2013	Student Assistant (PTT, SUMMERNET, JICA-RI/AIT projects)
Mar 2013 to Jul 2013	Intron Life Science Company
From 6 th Sep to 9 th Sep 2012	Internship at National Center for Aqua-seed in Central Vietnam – Research Institute for Aquaculture No.3
11 th Apr - 23 rd May 2011	Internship at Dong Nai Department of Science and Technology
6 th Aug 2010	Internship at National Breeding Center for Southern Freshwater Aquaculture
5 th Mar 2010	Internship at Tiger Beer Company

RESEARCH INTEREST

1. Development of emissions inventories for anthropogenic sources and biogenic sources.
2. Applications of air quality model and emission model for air quality management.
3. Indoor air quality monitoring and management.
4. Odor emission monitoring, modeling, and mitigation measures.
5. Health risk assessment.
6. Inverse modeling for emission inventory uncertainty assessment.
7. Assessment of air quality and climate co-benefits for Asian city.
8. Dioxin monitoring and analysis in Vietnam.
9. Dry and wet deposition for Greater Mekong Region (GMS).
10. Development of city's clean air action plan.
11. Source emission reduction strategies, planning, and management.

PROFESSIONAL SKILLS

1. Technical skills:
 - a) Statistical tools (SPSS), GIS.
 - b) Modeling system (IVE for traffic emission), PMF (for source emission), WRF (for meteorological simulation), CAMx (for air quality simulation), LEAP-IBC (integrated modelling and planning tool), CMB (Chemical Mass Balance)
 - c) OC/EC instrument, HPLC, odor monitor, basic chemistry instruments, etc.
 - d) Microorganism test, water, and wastewater test.
 - e) Satellite data (NASA, Himawari, Sentinel)
 - f) Cloud computing (Google Cloud Studio, HUPI)
 - g) Emission inventory tools (ABC-EIM)
 - h) Python, R
2. Soft skills:
 - a) Group work and leaderships
 - b) Good presentation skills
 - c) Good reading and writing academic works
 - d) Coordination and communication
3. English proficiency:
 - a) IELTS (overall: 7.0): Reading: 8.0; Writing: 6.5
4. Workshop/conference/training preparation and organization
5. Training
6. Project management, MEL, planning, cooperation, and communication
7. Project financial management
8. Project proposal preparation

PROJECT AND ACTIVITY INVOLVEMENT

Dr. Lai Nguyen Huy has handled 25 research projects in the capacity of Co-Principal Investigator (Co-PI), project manager/coordinator, and key resource personnel. Some are listed below:

Year	Name of project/activity	Sponsor / Consortium	Project description	Role
2025-2028	Climate Attribution of Wildfire Smoke Impacts on Priority Population Health in Southeast Asia and Australia (CANBREATHE)	Wellcome Trust PI: Prof. Nguyen Thi Kim Oanh	The project engages policymakers, Indigenous peoples and communities, citizen scientists and artists in the participatory co-design of innovative communication tools based on climate attribution science, health impact data, and the culture and lived experiences of communities suffering extreme or recurrent wildfires in Thailand, Laos, Indonesia and Australia.	Co-PI Project coordinator Key personnel for emission inventory development, and air quality modeling
2025-2027	Climate change adaptation to smoke haze for improved child health in Southeast Asia	e-Asia joint research PI: Prof. Nguyen Thi Kim Oanh	This interdisciplinary research project aims to assess the effectiveness of climate change adaptation interventions for strengthening community resilience to smoke haze in Southeast Asia, and increasing their adoption and implementation in the region to protect children's health. The project takes an integrated and multisectoral approach to manage the inter-related factors of adaptation interventions, such as early warning systems, clean air rooms, and targeted public health information, in culturally appropriate ways that promote their acceptance and implementation in local communities.	Co-PI Project coordinator Key personnel for emission inventory development, and air quality modeling
2025-2027	Deliver training programme for implementing decrees for reducing open waste burning and air pollution	UNEP through CCAC PI: Prof. Nguyen Thi Kim Oanh Budget: 70,825 USD	The objective of this project is to support Cambodia in effectively reducing municipal solid waste open burning and improving air quality in the country through an action plan and a training program, which will strengthen the enforcement capacity of stakeholders. The project team will merge its strengths in stakeholder engagement, capacity building, and science-based policy guidance with technical expertise in air quality and solid waste management and data analysis to effectively implement this project.	Co-PI Project coordinator Key personnel for emission inventory development, training programme, and development of action plan for reducing MSWOB
04-08/2025	Assessment of agricultural and forest burning emissions in Lao PDR	World Bank PI: Prof. Nguyen Thi Kim Oanh	This project aims specifically to analyze the air pollution status in Laos using multiple data sources (satellite data, emission inventory, etc.) for providing recommendations prioritizing emission control measures that lead to air quality improvements and simultaneously mitigate	Co-PI Project coordinator Key personnel for emission inventory development and analysis of satellite data

			emissions of climate forcers	
04-07/2024	East Asia Transboundary Air Pollution and Regional Air Quality Management	World Bank PI: Finish Meteorological Institute (FMI)	The objective of the project is to analyze and take stock of transboundary air quality and where possible air pollution dynamics between key countries in East Asia over the last year.	Assessment of Emissions Inventories of Air Pollution, transboundary pollution, and provide recommendations in Southeast Asia
03-12/2024	Stocktaking Methodological Assessment of Emissions Inventories and Source Apportionment of Air Pollution in Southeast Asia	UNEP/EANET PI: Dr. Verma Ram Lal	The major objective of the project is to understand emissions inventories and source apportionment in Southeast Asia and create a database of relevant emissions inventory and source apportionment studies and activities in Southeast Asian countries.	Assessment of Emissions Inventories of Air Pollution in Southeast Asia
2021-2024	Assessment of Soil Acid Deposition of the Pollutants from the Mae Moh Powerplant Number 4-7	NRTC, Thailand PI: Dr. Ekbordin Winijkul	This project is designed under the recommendation of the Environmental Health Impact Assessment that the Mae Moh Powerplant Number 4-7 needed to conduct the impact assessment of the soil condition degradation around the powerplant due to the emission from the operation of the powerplant	Acid deposition modeling Emission inventory QA/QC
2023-2025	Air quality improvement program in Thailand	AFD, France Egis, AIT CITEPA, and Airparif	The project aims to provide program management and transversal technical advisory, support to develop activities led by Pollution Control Department (PCD) of Ministry of Natural Resource and Environment (MoNRE), support to develop activities led by Office of Transport and Traffic Policy and planning (OTP) of Ministry of Transport (MoT), and support to develop activities led by Chiang Mai Municipalities & Province	Co-PI of AIT team Emission inventory development Overall project management.
2022	Smart Power Programs (SPPs) in Southeast Asia	USAID Deloitte, ADPC, AIT	AIT is a partner of this 5-year project. AIT tasks: SPP analysis and road map of power sector in SEA, emissions inventories for TPPs in focal SEA countries, impact assessment, and interventions analysis	Deputy AIT team leader. Lead the task on emission inventory and reports Overall project management.
2022-2026	Kathmandu Clean Air Program (K-CAP) or USAID Clean Air	USAID FHI 360, AIT, ENPHO, EPIC, One to Watch	The project promotes the application of data to improve evidence-based air pollution prevention and management in Nepal. The approach will support local governments, civil society organizations (CSOs), private sector companies and citizens to develop, test and scale locally identified solutions through the iterative collection and use of data	Overall project management Project manager Overall project technical tasks.
2020-2024	Nanoassisted	USAID	The proposed project aims to	Overall budget

	bioremediation of diffused dioxins in soil and sediment	AITCV/AIT, VLU, HCMUT	develop technologies to eventually eliminate dioxins using enriched indigenous microbes which have a high potential for application to convert the passive landfills into the active landfills. Laboratory and pilot-scale experiments will be conducted for this purpose. The findings of the project serve as the technical input for designing full-scale treatment suitable for the local conditions in Vietnam	project management Final workshop organization Coordination
2021-2022	Application of LEAP tool to beat air pollution for Pakistan	Clean Air Asia SEI, CACC	Application of LEAP tool to beat air pollution in Pakistan and building capacity for air quality managers in Pakistan	Coordinator of CAA team. Support technical on LEAP and emission inventory development
2021-2022	Development of emissions inventories for Baguio city and Paranaque city – the Philippines	Clean Air Asia	Development of emissions inventories for Baguio City and Paranaque City (the Philippines) and building capacity for air quality managers in the cities	Coordinator of the CAA team. Led the technical tasks of emission inventory development Training
2021-2022	Application of LEAP-IBC tool for Manila City – Asia Blue Skies Programme	3M Clean Air Asia	Application of LEAP tool to beat air pollution in Manila City and building capacity for air quality managers in Manila	Coordinator of the CAA team. Support technical on LEAP and emission inventory development.
2020	Improving Air Quality in Vietnam	World Bank AIT, HUS	The project developed emission inventory for three 3 provinces of Hanoi, Bac Ninh, and Hung Yen (Northern Vietnam) for air quality modeling and management purposes	Led the technical tasks of emission inventory development.
2020	Development of web-based emission inventory in Thailand	Pineapple Ltd. PCD, AIT	The project developed the emission inventory for Thailand in 2018, covering major anthropogenic and natural sources. The emission inventory data were then distributed into province data and uploaded to the website PCD for publicization.	In charge of emission inventory development for the waste sector.
2019-2020	Study of Carbon and Energy Balance of Different Sustainable Crop Residue Management in Asia	FAO AIT	The project aims to study the status and the carbon/energy balance of agricultural residue utilization in Asian countries. The analysis of scenarios of reduction in amount of agricultural residue subjected to open burning and the sustainable land management of alternative wetting and drying; and the winter legume plantation, in rice paddy were conducted.	Develop methodology framework of carbon and energy balance of different crop residue management in Asia. Led the project's final report and workshop organization.
2020	Plastic monitoring and assessment in the Lower Mekong Countries	RRC.AP (AIT)	The project aims to assess the current situation of plastic monitoring and management in the Lower Mekong Countries (LMC)	Collect and validate the data on plastic monitoring and

				management of countries in LMC.
2020-2023	Atmospheric deposition to large river basins and potential effects on the water environment	Sida SEI, AIT, NLU	DEPOSITION project used an integrated monitoring and modeling approach to assess the input of target compounds into the large water basins under the base case and emissions scenarios. The project was pioneering the effort to conduct measurements and compile a database to contribute to the new knowledge of the atmospheric deposition and effects on surface water quality and subsequent effects on local communities in the region.	Overall project management. Led technical activities of emissions inventory development. Organize national and international workshops.
2019	Capacity building program for professional air quality managers	UNEP/APCAP AIT	The project aims to build the capacity for Asian air quality managers. The training includes air quality managers in over 25 countries.	Organize 2 international workshops. In charge of a module on practicing using the emission inventory calculator.
2018-2022	Erasmus + Master on New Technologies Using Services" (MONTUS Erasmus +)	Erasmus 11 country partners in Europe and Asia		Overall project management Budget management Co-author of a training module on air quality management.
2015-2020	TORUS-Toward an Open Resources Upon Services: Cloud of environmental data	Erasmus 11 country partners in Europe and Asia		Overall project management Co-author of two book chapters on air quality modeling and emission inventory.
2018	Volunteer for the 5th Green Industry Conference for a Sustainable Future (3-5 of October 2018, Bangkok, Thailand) by UNIDO, MOI, and UNESCAP.			
2016-2017	Resource personnel for odor and H ₂ S monitoring and modeling projects at some sewage treatment plants in Hong Kong. These consultant projects combined both monitoring and modeling of odor and H ₂ S concentrations at some waste facilities in Hong Kong such as sewage treatment plants, sludge treatment, etc.			
2016-2017	Indoor PM _{2.5} levels at residential homes in four Chinese cities		The project analyzes the indoor PM _{2.5} and its associated components of BC, OC, heavy metal, and soluble ions samples collected at residential homes in four Chinese cities.	Design the sampling plan. In charge of monitor in Hongkong city Analyze EC/OC.
2015-2017	Atmospheric monitoring in the Bangkok area	JICA-Ri AIT, ACAP		Team support
2015-2016	Research for the development of a co-benefit action plan for Bandung	JICA AIT, IGES, MOFA	The project aims to conduct surveys for developing on-road traffic emissions in Bandung City and analyze a co-benefit action	Analyze and QA/QC local survey data. Run IVE modeling

	City, Indonesia: Traffic emission inventory	plan for the city.	for developing on- road traffic emissions in the city.
2013-2015	Satellite and Model Data use for Aerosols and Air Quality. Planning Workshop for Developing a Framework for Cooperation between IGAC Activities in Asia. Turning Rice Straw into Cooking Fuel for Air Quality and Climate Co-benefit in Selected GMS Countries. Ground-level Ozone in Bangkok Metropolitan Region by Advance Mathematical Modeling for Air Quality Management		Student assistant.

TEACHING AND TRAINING EXPERIENCE

1. Training key personnel for Section HYPLIT Modeling and Fire Hotspot in Module “Air quality management and transboundary air pollution in Southeast Asia” at Technical workshop and study visit on Transboundary Haze Pollution Management during 18-27 January 2024, organized by PCD and AIT.
2. Training key personnel for air quality modeling in Kathmandu, Nepal 11-12 October, 2023. Organized by FHI 360, AIT, and Tribhuvan University.
3. Training key personnel for CMB and IVE models training, and household survey case study in Kathmandu, Nepal 6-7th July 2023. Organized by FHI 360, AIT, and Tribhuvan University.
4. Training key personnel for emissions inventory training workshop in Kathmandu, Nepal, 6-7th April 2023. Organized by FHI 360, AIT, and Tribhuvan University.
5. Training personnel for emission inventory practices for Modules: Emissions inventories for governmental air quality managers in Paranaque
6. Training personnel for emission inventory practices for Modules: Emissions inventories for governmental air quality managers in Baguio City, Philippines. Organized by Clean Air Asia, Paranaque, and Baguio City Departments (2021).
7. Training personnel policy makers and governmental officers in Vietnam. Training modules: Emissions inventories and receptor modeling for governmental air quality managers in Vietnam. Organized by Live & Learn, NCEM (Hanoi, Vietnam) and USAID (2021).
8. Training personnel for emission inventory practices for UNEP/APCAP-Capacity project on “Capacity building program for professional air quality managers” training at AIT (2018).
9. Teaching and laboratory assistant for courses “Risk assessment” and “Indoor air pollution” at the Hong Kong Polytechnic University (2016 – 2017).

PUBLICATIONS

I. Journal Articles

1. Nguyen, Q. L., **Lai, H. N.**, Nguyen, H. T. M., Pham, L. L., Herdean, A., Poddar, N., Kuzhiumparambil, U., Nguyen, A., Pernice, M., Ralph, P. J., & Vo, P. H. N. (2025). Explainable artificial intelligence for predicting rare earth elements leaching from secondary resources. *Journal of Hazardous Materials*, 496, 139479. doi:<https://doi.org/10.1016/j.jhazmat.2025.139479>
2. Zusman, E., Akahoshi, K., Hanaoka, T., Malley, C.S., Wangwongwatana, S., Homyok, P., Pawarmart, I., Kim Oanh, N.T., **Huy, L.N.**, Kuylensstierna, J.C., et al., 2025. The truth about co-benefits: A multidimensional feasibility assessment for Thailand and beyond. *Environmental Research Communications* (accepted).
3. Akahoshi, K., Zusman, E., Hanaoka, T., Kim Oanh, N.T., **Huy, L.N.**, Wangwongwatana, S., Homyok, P., Malley, C.S., Hirayama, T., Goto, Y., et al., 2024. The prospects of controlling open burning of crop residues in Thailand: A quantitative assessment of implementation barriers and costs. *Atmosphere*, 15, 1309. <https://doi.org/10.3390/atmos15111309>
4. Kim Oanh, N. T., **Huy, L. N.**, Maneepatra, W., Winijkul, E., Giandomenico, A., Tantrakarnapa, K., Co, H. X., Cuong, D. M., Tsou, M. C. M., Hien, T. T., Chi, N. D. C. T., Ngan, T. A., Lung, S. C. C. (2024). Comparative analysis of PM_{2.5} levels in various microenvironments associated with common cooking practices in selected Asian countries. *Air Quality, Atmosphere & Health*. <https://doi.org/10.1007/s11869-024-01615-1>.
5. Vo, P. H. N., Ky Le, G., **Huy, L. N.**, Zheng, L., Chaiwong, C., Nguyen, N. N., Nguyen, H. T. M., Ralph, P. J., Kuzhiumparambil, U., Soroosh, D., Toft, S., Madsen, C., Kim, M., Fenstermacher, J., Hai, H. T. N., Duan, H., & Tscharke, B. (2024). Occurrence, Spatiotemporal Trends, Fate, and Treatment Technologies for Microplastics and Organic Contaminants In Biosolids: A Review. *Journal of Hazardous Materials*, 133471. doi:<https://doi.org/10.1016/j.jhazmat.2024.133471>
6. Kim Oanh, N.T. and **Huy, L.N.** (2024). Atmospheric emissions from electricity generation in Southeast Asia: development trend and policy responses. *Current Pollution Reports*. <https://doi.org/10.1007/s40726-023-00289-0>
7. Vo, P. H. N., Danaee, S., Hai, H. T. N., **Huy, L. N.**, Nguyen, T. A. H., Nguyen, Hong, T. M., Kuzhiumparambil, U., Kim, M., Nghiem, L. D., Ralph, P. J. (2023). Biomining for sustainable recovery of rare earth elements from mining waste: A comprehensive review. *Science of The Total Environment*, 168210. <https://doi.org/10.1016/j.scitotenv.2023.168210>
8. **Huy, L.N.**, Kim Oanh, N.T., Huong, C.T.T., Huyen, T.T., 2022. Analysis of atmospheric emissions associated with on-road and inland waterway transport in Vietnam: Past, current, and future control scenarios. *Atmospheric Pollution Research*, 101810.
9. Makkonen, U., Vestenius, M., **Huy, L.N.**, et al., 2023. Chemical composition and potential sources of PM_{2.5} in Hanoi. *Atmospheric Environment*, 299, 119650, <https://doi.org/10.1016/j.atmosenv.2023.119650>
10. Huyen, T.T., Kim Oanh, N.T., **Huy, L.N.**, Winijkul, E., Ha Chi, N.N., 2022. Emission inventory of air pollutants and greenhouse gases from shipping activities in Ha Long Bay, Vietnam. *Environmental Technology and Innovation*, 27, 102507.
11. **Huy, L.N.**, Winijkul, E., Kim Oanh, N.T., 2021. Assessment of Emissions from Residential Combustion in Southeast Asia and Implications for Climate Forcing Potential. *Science of the Total Environment*, 785, 147311.
12. **Huy, L. N.**, Kim Oanh, N.T.K., Phuc, N.H., Nhung, C.P., 2021. Survey-based inventory for atmospheric emissions from residential combustion in Vietnam. *Environmental Science and Pollution Research*, 28, 10678–10695.
13. **Huy, L.N.**, Kim Oanh, N.T., 2020. Emission control for volatile organic compounds from gasoline stations and implication on ozone-forming potential. *Atmospheric Pollution Research*, 11(6), 87–98.
14. **Huy, L.N.**, Kim Oanh, N.T., Htut, T.T., Hlaing, O.M.T., 2020. Emission inventory for on-road traffic fleets in Greater Yangon, Myanmar. *Atmospheric Pollution Research*, 11(4), 702–713.

15. Perera, G.B.S., Manthilake, M.M.I.D., Sugathapala, A.G.T., **Huy, L.N.**, Lee, S.C., 2019. NO_x-VOC-O₃ sensitivity in urban environments of Sri Lanka. *Asian Journal of Atmospheric Environment*, 13(1), 62-72.
16. Kim Oanh, N.T., **Huy, L.N.**, Permadi, D.A., Zusman, E., Nakano, R., Nugroho, B., Lestari, P., Sofyan, A., 2018. Assessment of urban passenger fleet emissions to quantify climate and air quality co-benefits resulting from potential interventions. *Carbon Management*, 9(4), 367-381.
17. **Huy, L.N.**, Lee, S.-C., Zhang, Z., 2018. Human cancer risk estimation for 1,3-butadiene: An assessment of personal exposure and different microenvironments. *Science of the Total Environment*, 616-617, 1599-1611.
18. **Huy, L.N.**, Kim Oanh, N.T., 2017. Assessment of national emissions of air pollutants and climate forcers from thermal power plants and industrial activities in Vietnam. *Atmospheric Pollution Research*, 8(3), 503-513.
19. Danh, N.T., **Huy, L.N.**, Kim Oanh, N.T., 2016. Assessment of rice yield loss due to exposure to ozone pollution in Southern Vietnam. *Science of the Total Environment*, 556-557, 1069-1079.
20. Do, K.P., Nguyen, B.T., Nguyen, X.T., Bui, Q.H., Tran, N.L., Nguyen, T.N.T., Vuong, V.Q., **Huy, L.N.**, Le, T.H., 2015. Spatial interpolation and assimilation methods for satellite and ground meteorological data in Vietnam. *Journal of Information Processing Systems*, 11(4), 556-572.

II. Conferences/Workshops

1. Huy, L.N., Kim Oanh, N.T., 2025. Assessment of Biomass Open Burning Emissions in Southeast Asia. Poster presentation at 6th Asian Air Pollution and Climate Change (AAPC), 20 – 22 November 2025, Tsukuba, Japan.
2. **Huy, L.N.**, Winijkul, E., Kim Oanh, N.T., 2022. Development of Black Carbon Emissions for Southeast Asia. Presented at 2022 International Symposium on Clean Environment and Planetary Health in Asia (CEPHA 2022). March 26th-28th, 2022, Bangkok, Thailand.
3. **Huy, L.N.**, Winijkul, E., Kim Oanh, N.T., 2020. Emission Inventory of Air Toxic Pollutants from Household Solid Waste Open Burning in Vietnam. Poster presentation at 19th GEIA Conference, Santiago, Chile.
4. **Huy, L.N.**, Lee, S.-C., 2017. Indoor air pollution in residential homes of Hong Kong. Presented at *Health Building Asia 2017*, Tainan, Taiwan.
5. **Huy, L.N.**, Kim Oanh, N.T., 2015. Development of national industrial emission inventory for Vietnam. Presented at 108th Air and Waste Management Association Annual Conference and Exhibition (ACE 2015), North Carolina, USA.

III. Books/Book chapters

1. Moukhtar, S., Andre, J.-M., Bedrune, Q., Braish, T., Cuniasse, B., Hercule, J., Kamar, R., Mazin, V., Mercier, A., Robert, C., **Huy, L.N.**, Winijkul, E., Kim Oanh, N.T. 2024. Air Quality Improvement Program in Thailand: Sources of Air Pollution in Chiang Mai in 2022 - Main Sources and Measures to Mitigate Air Pollution.
2. Kim Oanh, N.T., **Huy, L.N.**, Verma, R.L., et al. 2023. State of Air in Nepal with focus in Kathmandu Valley. DOI: 10.5281/zenodo.13335444
3. Kim Oanh, N.T., **Huy, L.N.**, Winijkul, E., Thanh, N.T.N., Hung, N.T.Q., 2022. Chapter 12. Integrated Air Quality Management. <https://cloud-montus.com/foundations-of-environmental-sciences/>. <https://zenodo.org/record/7533133#.Y-39dsdBxD8>
4. Kim Oanh, N.T., Ha Chi, N.N., **Huy, L.N.**, Permadi, D.A., 2019. Chapter 29. Atmospheric modelling with focus on management of input/output data and potential of cloud computing applications. In *TORUS 3 – Toward an Open Resource Using Services*, Lafly, D. (ed.), Wiley: New York.
5. Kim Oanh, N.T., **Huy, L.N.**, Permadi, D.A., Ha Chi, N.N., Sothea, K., Chitapornpan, S., Kanabkaew, T., Ratanarat, J., Sichum, S., 2019. Chapter 28: Emission inventories for air pollutants and greenhouse gases with emphasis on data management in cloud. In *TORUS 3 – Toward an Open Resource Using Services*, Lafly, D. (ed.), Wiley: New York

IV. Technical reports

1. Kim Oanh, N.T., **Huy, L.N.**, Dang, H.Q., Oo, K.T. 2025. Assessment of agricultural and forest burning emissions in Lao PDR. Report submitted to the World Bank Group
2. Moukhtar, S., Andre, J.-M., Bedrune, Q., Braish, T., Cuniasse, B., Hercule, J., Kamar, R., Mazin, V., Mercier, A., Robert, C., **Huy, L.N.**, Winijkul, E., Kim Oanh, N.T. 2024. Comprehensive Inventory Methodology Report for Chiang Mai Province, Thailand (2022). Report of Air Quality Improvement Program in Thailand. Submitted to AFD.
3. Andre, J.-M., Bedrune, Q., Braish, T., Cuniasse, B., Hercule, J., Kamar, R., Mazin, V., Mercier, A., Moukhtar, S., Robert, C., **Huy, L.N.**, Winijkul, E., Kim Oanh, N.T. 2024. Sources of Air Pollution in Chiang Mai in 2022 – Main sources and measures to mitigate air pollution. Report of Air Quality Improvement Program in Thailand. Submitted to AFD.
4. Kim Oanh, N.T., **Huy, L.N.**, Huyen, T.T., Borongan, G., Kyaw, M., Verma, R.L. (2024). Stocktaking and Methodological Assessment of Emission Inventories and Source Apportionment Studies in Southeast Asia: Meta-analysis and Database. Report= submitted to UNEP/EANET
5. **Huy, L.N.**, Kim Oanh, N.T. 2024. Background Study on Two-Wheeler Delivery Companies (Last-mile Delivery Services) Towards Inclusive, Low-carbon Transport Transformation in Thailand. Report submitted to ICSC.
6. Loven, K., Sundstrom, A.-M., Linqvist, H., Suhonen, E., Kim Oanh, N.T., **Huy, L.N.**, Huyen, T.T. (2024). East Asia Transboundary Air Pollution and Regional Air Quality Management. Report submitted to the World Bank Group.
7. Kim Oanh, N.T., **Huy, L.N.** 2023. The development of biomass fuels and stoves in Thailand. Technical report submitted to BUCT hybrid workshop (18-20th October 2023), Beijing, China.
8. Kim Oanh, N.T., **Huy, L.N.**, Swe, I.M., Tuan, L.Q., et al. 2023. Atmospheric deposition to large river basins and potential effects on the water environment. Final report submitted to SEI-SUMERNET 4 All.
9. Kim Oanh, N.T., **Huy, L.N.**, Oo, K.T., Swe, I.M., Tung, H.D., Trung, N.Q., Permadi, D.A., 2022. Policy Landscape Impacting Air Quality in the Power Sector of Southeast Asia. Technical report submitted to USAID Southeast Asia's Smart Power Program (SPP) funded by USAID.
10. Kim Oanh, N.T., **Huy, L.N.**, Tung, H.D., Trung, N.Q., Permadi, D.A., 2022. Assessment of the current situation of air pollution from the power sector in Southeast Asia. Technical report submitted to USAID Southeast Asia's Smart Power Program (SPP) funded by USAID.
11. Kim Oanh, N.T., **Huy, L.N.**, Ha Chi, N.N. 2021. Emissions inventories in Hanoi, Bac Ninh, and Hung Yen in 2015 and 2019. A technical report as a part of project funded by World Bank.
12. Kim Oanh, N.T., **Huy, L.N.** 2021. PM_{2.5} emissions inventories in Vietnam and two populous cities in Vietnam. Technical report as a part of State of Air Pollution in Vietnam, a multi-source analysis by USAID, Live & Learn, and VNU.
13. Winijkul, E., Shrestha, R.P., **Huy, L.N.**, et al., 2021. Study on Carbon and Energy Balance of Different Crop Residue Management Practices in Asia. Technical report submitted to FAO - Food and Agriculture Organization of the United Nations.
14. Kim Oanh, N.T., **Huy, L.N.**, Patdu, M.K., Ha Chi., N.N., 2018b. Part 2 training on “capacity building programme for national air quality managers”. Technical report submitted to UNEP - UN Environment Programme.
15. Kim Oanh, N.T., **Huy, L.N.**, Patdu, M.K., Ha Chi., N.N., 2018a. Part 1 training on “capacity building programme for national air quality managers”. Technical report submitted to UNEP - UN Environment Programme.
16. Kim Oanh, N.T., **Huy, L.N.**, Permadi, D.A., 2016. On-road traffic emission inventory using international vehicle emission model for Bandung city, Indonesia. Technical report submitted to IGES and MoIJ.

V. Development and Introduction of Pedagogical Techniques

1. Kim Oanh, N.T., **Huy, L.N.**, Verma, R.L., 2023. Training program for air quality practitioners in Nepal. Topic: Emissions inventories in Integrated Air Quality Management.
2. Kim Oanh, N.T., **Huy, L.N.**, Winijkul, E., Thanh, N.T.N., & Hung, N.T.Q., 2022. Insights 12. Cookbook on Integrated Air Quality Management – within MONTUS project. Integrated Air Quality Management - Pedagogical resource available at <https://cloud-montus.com/foundations-of-environmental-sciences/>

V. Research webpages

1. Research Gate: <https://www.researchgate.net/profile/Lai-Huy-2>

VI. SERVICES

I. Reviewing service

1. Reviewer for the journals of Environmental Science & Technology, Science of Total Environment, Atmospheric Environment, Journal of Environmental Management, Aerosol and Air Quality Research, Air Pollution Research.
2. Guest Editor of Special Issue “Health Impact of Indoor Air Pollution” of Indoor Air

VII. AWARDS

Year	Description
January, 2018	PhD scholarship by AIT Silver and AIT fellowship scholarship, 2018 – 2021.
August, 2017	ISIAQ Student Conference Support Award winner for the Healthy Buildings 2017 Asia conference.
June,	2015 Student Paper Award at West Coast Section of the Air & Waste Management
May, 2015	The Robert B. Banks prize for the most outstanding academic performance in the field of Environmental Engineering and Management at AIT.
June, 2014	Internship scholarship and best student presentation at K-water Company (Daejeon, Korea)
2013-2015	Master scholarship by DAAD (Germany and AIT fellowship scholarship)
2008-2012	Scholarship of University Entrance Examination 2008 Scholarship of international university

VIII. MEMBERSHIP IN PROFESSIONAL SOCIETIES:

- The Health Investigation and Air Sensing for Asian Pollution (Hi-ASAP)
- Healthy Environments And Lives (HEAL) National Research Network

Signature:

A handwritten signature in black ink, consisting of a stylized 'L' followed by a cursive 'N' and 'H', with a long horizontal stroke extending to the right.

LAI NGUYEN HUY

Date: December 4, 2025